COMPLETE FABRICATION DESIGN - EXCLUDING DOME

Tubular Daylight Tube System

Build-oriented drawings and specifications for the complete remaining assembly: reflective tube, modular joints, dome interface, optional elbow, curb adapter, supports, insulation and vapor control, ceiling plenum, trim ring, and diffuser.

DESIGN BASIS	CONFIGURATION	STATUS	UNITS
Existing dome collar ID 356	Four 500 mm modules	Prototype - Rev A	mm unless noted

Scope and baseline

This package starts at the existing dome's lower collar and ends at the room-side diffuser. The baseline is a straight, 2,000 mm cut-length tube stack. Because 90 mm enters the dome collar and 75 mm enters the ceiling plenum, the resulting building-cavity reach is approximately 1,835 mm. The length formula in Sheet 1 allows any other site distance to be fabricated.

PARAMETER	DESIGN VALUE	CONTROL
Tube	Dia. 355 OD x 0.50 wall	Fits existing Dia. 356 ID dome collar
Optical material	Enhanced specular aluminum	Total reflection target 98 percent
Standard module	500 long	Cut blank 1130 x 500
Joint	100 split external sleeve	Butt tube ends; no internal fasteners
Supports	At <= 1000 spacing	Two M6 rods to verified structure
Thermal envelope	25 closed-cell insulation	Continuous exterior vapor seal
Ceiling opening	Dia. 400 +/-2	Trim OD 450; clear aperture Dia. 380

CRITICAL LIMIT
The roof curb and cap shown are a generic prototype interface, not a universal roof flashing. A qualified roofing and structural professional must design the actual roof opening, curb attachment, wind resistance, drainage, flashing, fire requirements, and weather seal for the building and roof system.

Drawing index

Sheet 1 - overall assembly; Sheet 2 - straight tube module; Sheet 3 - joints and interfaces; Sheet 4 - optional elbow and supports; Sheet 5 - curb-mounted roof adapter; Sheet 6 - ceiling plenum and diffuser.

ENGINEERING DRAWING - SHEET 1 OF 6

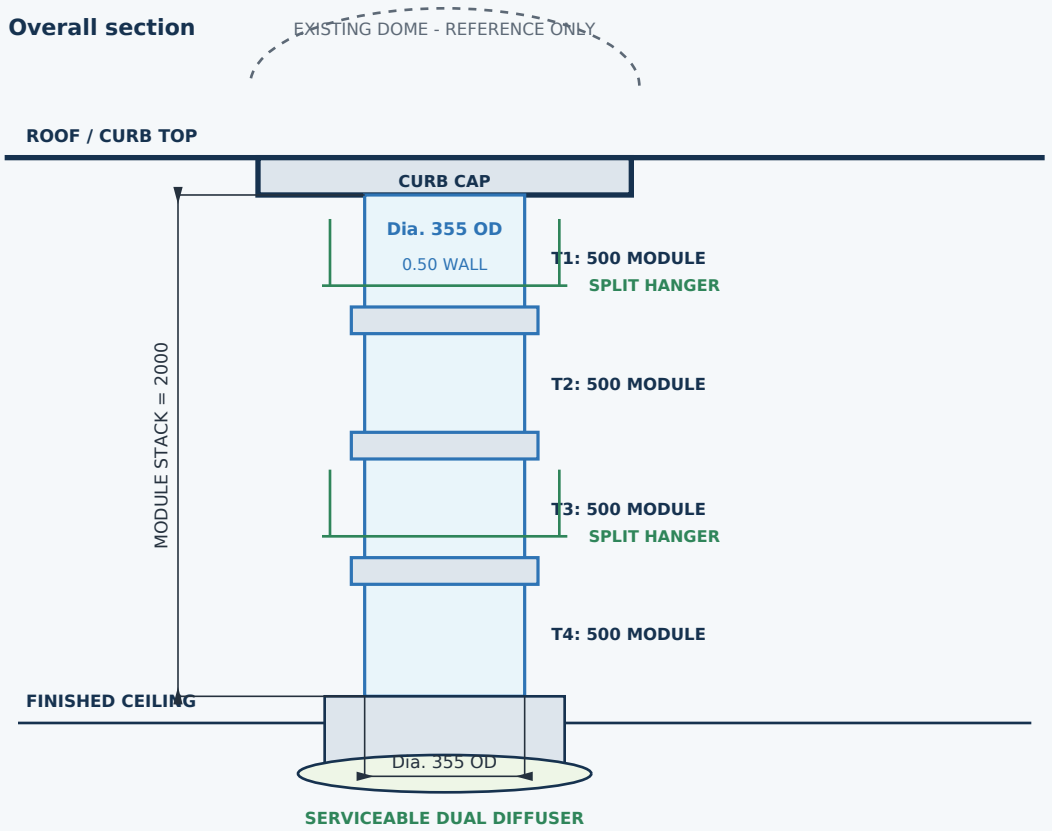
Overall assembly and controlling dimensions

COMPLETE TUBE SYSTEM - STRAIGHT BASELINE ASSEMBLY

SHEET 1 / 6

PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

A1 - Overall section



A2 - Controlling dimensions

Baseline: four 500 mm straight modules. The usable building-cavity reach is reduced by the 90 mm dome-collar insertion and 75 mm ceiling-plenum insertion.

Dome collar ID	**356.0**
Tube outside	**355.0 +/-0.5**
Tube inside	**approx. 354.0**
Wall	**0.50 +/-0.02**
Module length	**500.0 +/-0.5**
Stack cut length	**2000**
Installed reach	**1835**
Joint sleeve	**100 wide**
Insulation OD	**approx. 405**
Rough opening	**450 x 450 min.**

Length rule: Required total tube cut length = measured distance from dome-collar underside to plenum top + 165 mm. Divide into sections no longer than 600 mm for hand rolling and handling.

The baseline uses four 500 mm reflective modules. Use the length rule on the sheet to adapt the run to the actual roof-to-ceiling measurement.

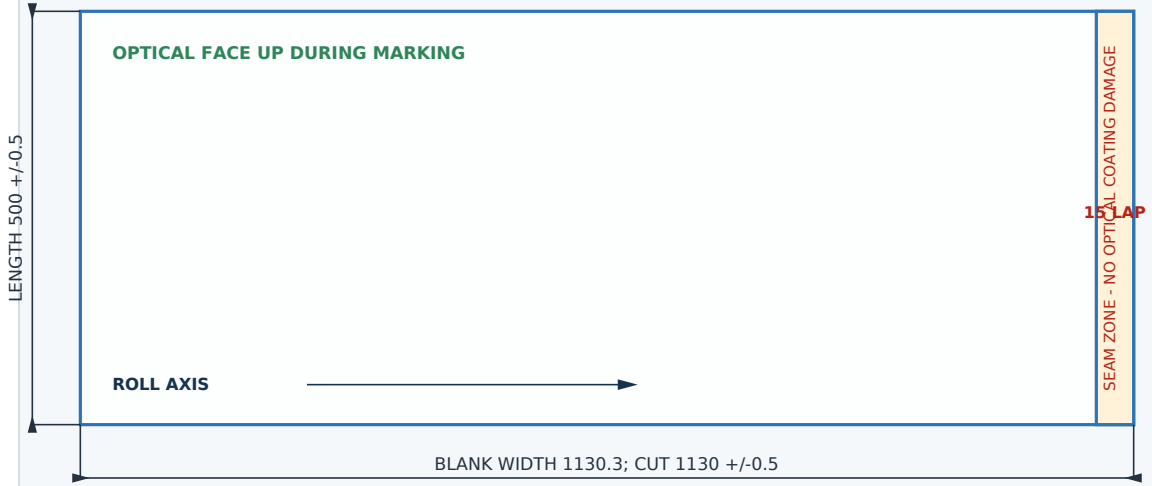
Reflective straight tube module and developed blank

REFLECTIVE STRAIGHT TUBE MODULE

SHEET 2 / 6

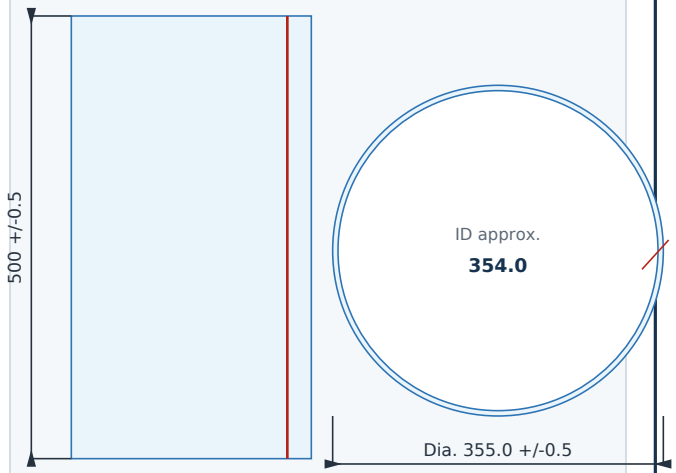
PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

B1 - Developed blank



Material: enhanced specular aluminum, 0.50 mm. Total light reflectance target 98 percent. Keep protective film on during cutting and rolling; remove only after the tube is round and all burrs are removed.

B2 - Rolled module and tolerances



- End squareness <= 0.50
- Roundness: max - min dia. <= 1.0
- No dents > 0.30 deep
- Inside seam step <= 0.30
- Lap edge faces toward diffuser
- No rivet heads in optical bore

This sheet gives the flat blank, formed diameter, seam direction, and inspection tolerances for each repeatable module.

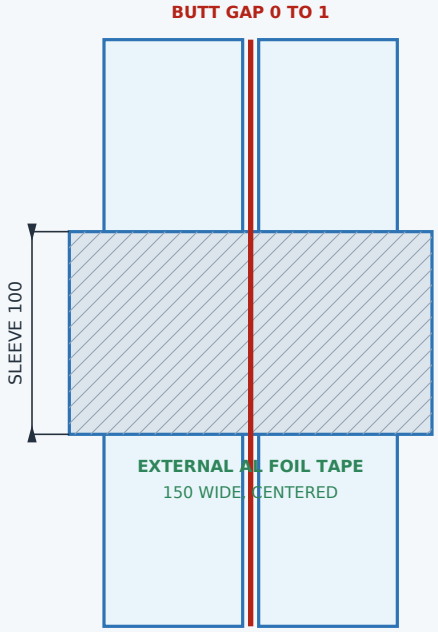
External joints and dome/plenum socket interfaces

JOINTS, DOME INTERFACE, AND LOWER SOCKET

SHEET 3 / 6

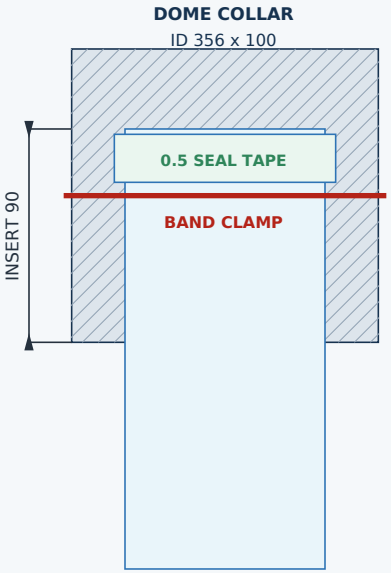
PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

C1 - Straight butt joint



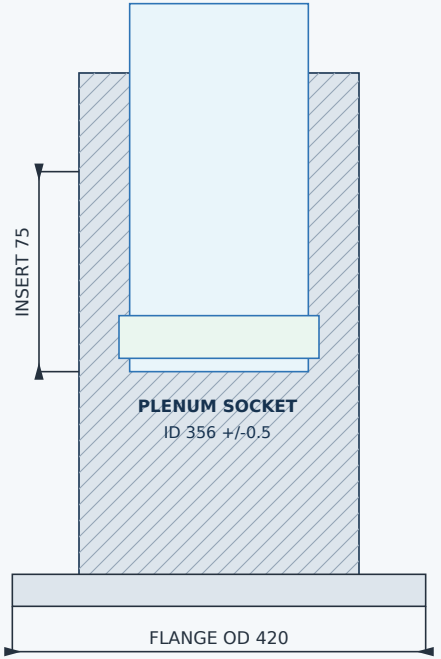
Split sleeve: 5052-H32 aluminum, 1.0 thick x 100 wide. Roll to 356.0 bore. Developed strip 1112 x 100, leaving 8 to 12 mm clamp gap. Add two 30 x 100 x 1.5 clamp ears and two M5 bolts.

C2 - Existing dome collar interface



Keep 10 mm clearance below the collar top. Seal with compatible 0.5 x 25 butyl/EPDM tape and an external stainless band clamp. Do not drill the optical bore.

C3 - Ceiling plenum socket



Seal with 0.5 x 25 compatible tape. Secure using an external band clamp. Support the plenum from framing; the diffuser trim is not structural.

All connections remain external to the optical bore. The tube ends butt together and are sealed outside the light path.

Optional fixed 30-degree elbow and structural supports

OPTIONAL 30-DEG ELBOW AND STRUCTURAL SUPPORTS

SHEET 4 / 6

PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

D1 - Fixed 30-deg miter elbow

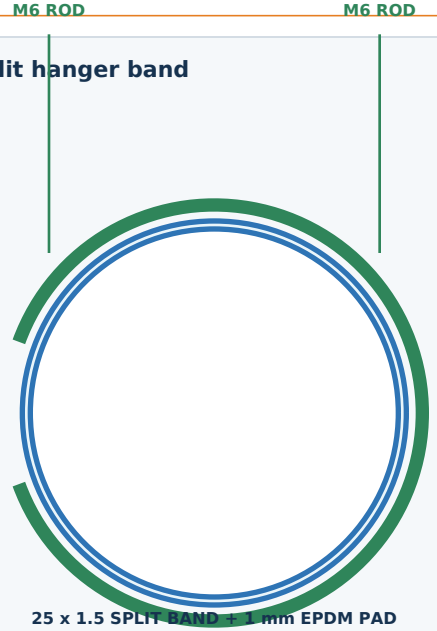


MITER TEMPLATE OFFSET FROM CENTERLINE

Angle	Offset
180 deg	**-47.6 mm**
150 deg	**-41.2 mm**
120 deg	**-23.8 mm**
90 deg	**+0.0 mm**
60 deg	**+23.8 mm**
30 deg	**+41.2 mm**
0 deg	**+47.6 mm**

Cut both tube ends at 15 deg and mirror them. Across a 355 mm tube, the long-to-short cut difference is 95.1 mm. Butt the edges with gap <= 1.5 mm. Bridge externally with four 20 x 120 x 0.8 straps at 90 deg, then seal with 100 mm aluminum foil tape. This fixed elbow is optional; every bend reduces delivered light.

D2 - Split hanger band



Support spacing: <= 1000 mm for the prototype and within 200 mm of each elbow, roof transition, and ceiling plenum. Attach rods only to verified structure. Do not hang the tube from the dome, curb cap, drywall, or diffuser ring. Allow axial thermal movement at one support in each run.

Use the elbow only where obstruction avoidance is necessary. Keep the run as straight and short as practical for best light delivery.

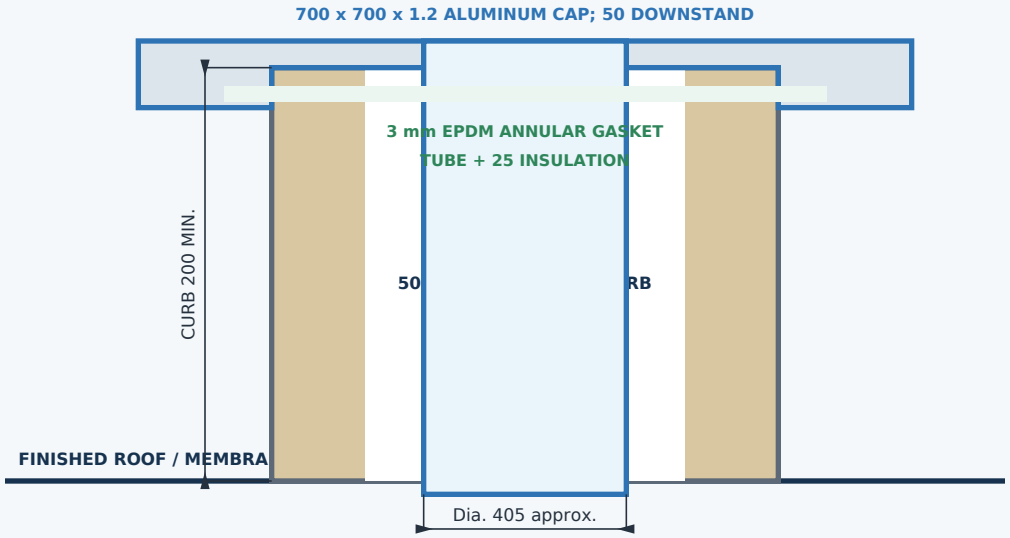
Generic curb-mounted roof adapter

GENERIC CURB-MOUNTED ROOF INTERFACE

SHEET 5 / 6

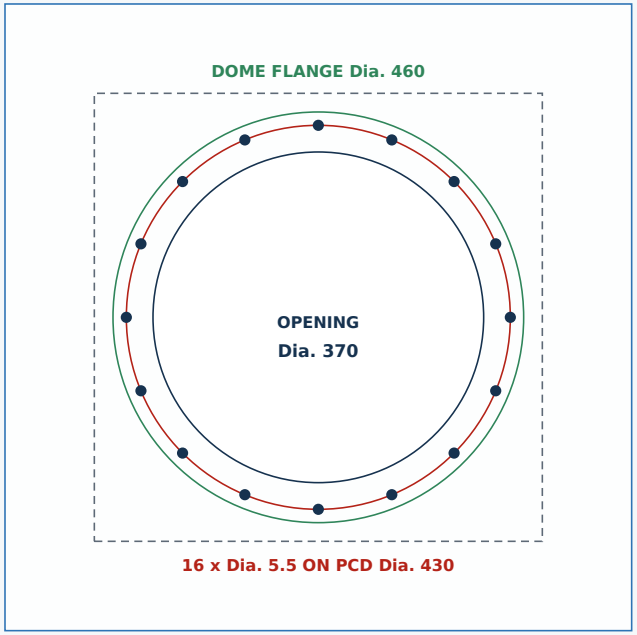
PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

E1 - Section through curb



The cap is a dome mounting adapter, not a universal flashing. The roofing contractor must integrate membrane, shingles, metal panels, drainage, ice/water protection, curb attachment, and counterflashing for the actual roof system.

E2 - Cap plate plan



CAP PLATE 700 x 700

Cap central flat zone Dia. 500 minimum. Opening Dia. 370. Backing ring below cap: OD 456, ID 374, 3.0 aluminum, 16 holes Dia. 5.5 on PCD Dia. 430.

The cap plate provides the dimensional interface to the existing dome. Site-specific curb, flashing, and attachment remain the responsibility of qualified building professionals.

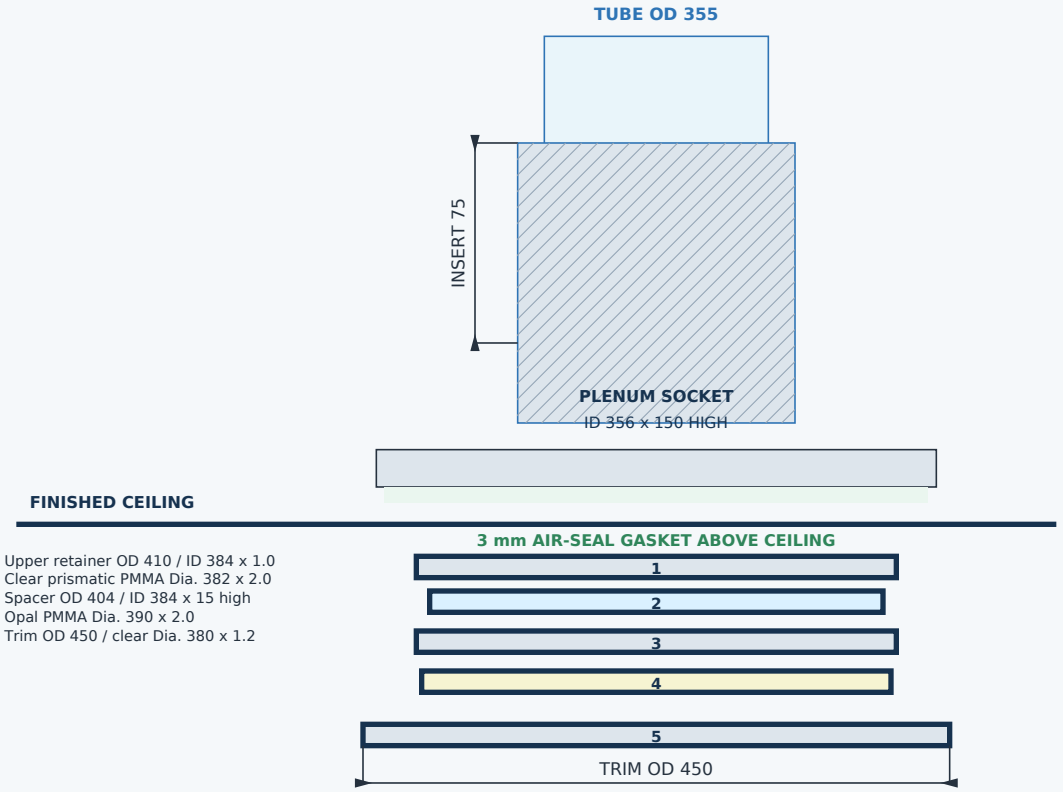
Ceiling plenum, air seal, trim ring, and diffuser

CEILING PLENUM, AIR SEAL, TRIM, AND DUAL DIFFUSER

SHEET 6 / 6

PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

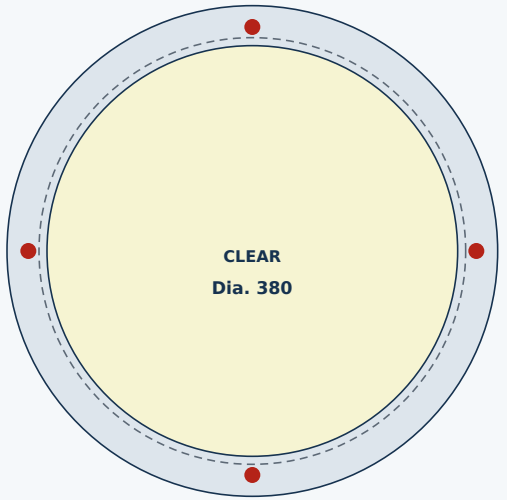
F1 - Exploded section



1 Upper retainer OD 410 / ID 384 x 1.0
2 Clear prismatic PMMA Dia. 382 x 2.0
3 Spacer OD 404 / ID 384 x 15 high
4 Opal PMMA Dia. 390 x 2.0
5 Trim OD 450 / clear Dia. 380 x 1.2

F2 - Plan and service dimensions

CEILING ROUGH OPENING Dia. 400 +/-2



4 SPRING CLIPS AT 90 deg

If exact diffuser transmission is unavailable, prototype 60, 70, and 80 percent opal sheets and choose the lowest-glare option that still meets the room illuminance target.

The diffuser is removable from below for cleaning and optical trials. The plenum must be supported independently from the ceiling finish.

1. Bill of materials and cut list

ITEM	PART	MATERIAL	QTY	CUT / FINISHED SIZE	CRITICAL REQUIREMENT
T1-T4	Straight tube	MIRO-SILVER 27 or equal enhanced specular Al	4	Blank 1130 x 500 x 0.50; formed Dia. 355	Optical face inward; total reflection target 98 percent
J1-J3	Joint sleeve	5052-H32 Al	3	1112 x 100 x 1.0; roll to Dia. 356 bore	8 to 12 clamp gap; no internal hardware
J4	Dome seal	Compatible butyl/EPDM tape	1	0.5 x 25 x 1120	Continuous, compressed; exterior only
J5	Foil joint tape	Outdoor aluminum foil tape	4	150 wide x approx. 1120 each	Seal each butt joint and lower socket
H1-H3	Hanger band	1.5 Al or stainless + EPDM pad	3	25 wide; roll to tube OD	Two M6 rods; structure verified
R1	Curb cap	5052-H32 Al	1	700 x 700 x 1.2; opening Dia. 370	Central flat zone Dia. 500
R2	Backing ring	5052-H32 Al	1	OD 456 / ID 374 x 3.0	16 holes Dia. 5.5 on PCD Dia. 430
P1	Plenum socket	5052-H32 Al	1	ID 356 x 150 x 0.8; flange OD 420	Tube insertion 75
D1	Upper diffuser	Clear microprismatic PMMA	1	Dia. 382 x 2.0	Deburr and retain mechanically
D2	Lower diffuser	Opal PMMA	1	Dia. 390 x 2.0	Trial 65 to 75 percent transmittance
D3	Trim ring	1.2 Al or code-compliant polymer	1	OD 450 / clear Dia. 380	Four service clips at 90 deg
I1	Tube insulation	Code-compliant closed-cell foam	As req.	25 thick	Continuous exterior vapor seal

MATERIAL CONTROL

Do not substitute ordinary mill-finish aluminum for the optical tube. The design depends on a protected enhanced-specular surface. Alanod MIRO-SILVER 27 is available at 0.40 to 0.50 mm and lists total light reflection of 98 percent; an equivalent material must provide documented optical performance and be formable without surface cracking.

2. Fabrication and assembly sequence

STEP	OPERATION	DETAIL / ACCEPTANCE
1	Measure site	Measure from the existing dome-collar underside to the ceiling-plenum top along the intended centerline. Add 165 mm for interface insertions.
2	Plan modules	Divide required cut length into sections <= 600. Keep joints clear of framing, elbows, and hanger locations.
3	Cut optical sheet	Cut blanks with protective film intact. Use shears or a guarded fine-tooth saw. Deburr the back edge without scratching the optical face.
4	Roll cylinders	Roll with optical face inward to Dia. 355. Form a 15 mm longitudinal lap; keep the inner edge step <= 0.30.
5	Close seam	Use compatible high-strength acrylic transfer tape in the lap and exterior foil tape. Do not put rivet heads inside the optical bore.
6	Make sleeves	Roll each 1112 x 100 x 1.0 strip to a 356 bore. Add clamp ears and M5 bolts; trial-fit without crushing the tube.
7	Build roof adapter	Fabricate the cap and backing ring, then have the site-specific curb, attachment, and flashing reviewed before opening the roof.
8	Build plenum	Roll the Dia. 356 socket, form the OD 420 flange, add two structural brackets, gasket, trim, and removable lenses.
9	Dry assemble	Assemble on the floor. Confirm roundness, bore continuity, joint gaps, diffuser removal, and overall centerline length.
10	Install and support	Install roof interface, tube modules, hangers, insulation and vapor seal, then plenum and diffuser. Remove all protective films last.

Assembly controls

- Orient every longitudinal lap edge toward the diffuser so incoming rays meet the smooth side first.
- Align all modules within 2 mm over the full run; avoid visible steps at joints.
- Tighten split sleeves only until the tube cannot rotate by hand. Crushing or ovalizing the bore reduces performance.
- Seal every exterior tube joint airtight before applying insulation. Continuously seal the insulation vapor layer as required by the building climate and code.
- Keep the tube isolated from copper, wet treated wood, masonry salts, and incompatible sealants using approved separation materials.
- Provide access to remove the diffuser and clean the tube without dismantling the roof interface.

CONDENSATION AND FIRE

The 25 mm insulation shown is a prototype starting point, not a universal energy or condensation solution. Determine insulation thickness, vapor-retarder location, fire rating, smoke development, and penetration detailing for the actual climate, roof assembly, and occupancy.

3. Verification, optical commissioning, and limits

CHECK	METHOD	INITIAL PROTOTYPE TARGET
Tube diameter	Measure at 0, 45, 90, and 135 deg at both ends	355.0 +/-0.5; max-min <= 1.0
End squareness	Flat plate and feeler gauge	<= 0.50 gap
Joint alignment	Straightedge across each sleeve and bore inspection	Axis offset <= 1.0; butt gap 0 to 1.0
Air leakage	Low-pressure smoke or tracer test before insulation	No visible leakage at seams, joints, plenum, or dome collar
Water control	Qualified roof spray test including upslope and side laps	No water inside curb, insulation, ceiling plenum, or room
Support	Inspect rods, brackets, clamps, and thermal movement	No load on dome, flashing, drywall, or diffuser
Optical throughput	Compare outdoor horizontal illuminance with room-side diffuser illuminance or flux	Record clear-day and overcast baselines; no joint shadow bands
Glare and diffusion	View at normal room positions under low and high sun	No exposed bright tube image; change opal transmission if needed
Thermal cycle	Minimum 10 hot/cold cycles with inspection	No tape release, condensation staining, corrosion, distortion, or seal movement

Optical configuration rules

- Keep the run straight whenever possible. Use the optional 30-deg elbow only to clear an obstruction and avoid more than two elbows.
- Keep tube interior dry, dust-free, and free of fingerprints. Clean only with the optical-material supplier's approved method.
- Do not paint the optical surface or cover it with ordinary aluminum tape except for a narrow repair approved during prototype testing.
- Longer runs are possible, but the prototype should be photometrically tested at the actual length and bend count before the design is released.

Primary technical references

- VELUX USA, Pitched TMR rigid Sun Tunnel product data: 14-inch system, >98 percent total reflectance, 45-deg adjustable elbow benchmark, and installation-length information. https://pro.veluxusa.com/products/sun-tunnels/rigid-sun-tunnel/tmr

- Alanod, MIRO-SILVER 27 product data sheet, dated 2025-03-03: 0.20 to 0.50 mm material and 98 percent total light reflection. https://alanod.com/_Resources/Persistent/7/2/f/e/72feb130d99a2dd1b4dbf3f2fe603ca72935d5a2/MIRO-SILVER%2027-EN.pdf

- Solatube SolaMaster specification: coordinate roof flashing with insulation, air barriers, vapor retarders, and weather-tight installation. https://solatube.com/wp-content/uploads/2021/02/Arc_Spec_08625_SolaMaster_750_DS.pdf

- PLEXIGLAS Satinice product information: matte acrylic diffusion options for lighting applications. https://www.plexiglas.de/en/products/plexiglas/plexiglas-satinice

- VELUX USA skylight testing information: structural and performance testing context for commercial roof products. https://pro.veluxusa.com/tools/architects/skylight-testing-data

RELEASE AUTHORITY

This package is suitable for prototype fabrication and optical development. It is not a certified skylight, roofing, firestop, structural, or energy-code design. Permanent installation requires review and approval by qualified professionals and any required laboratory or authority having jurisdiction.